Final Report
Recipe Characteristics and User Ratings Analysis

Ecehan Kalkan

Abstract

This project analyzes how recipe characteristics relate to user ratings using the Food.com Recipes and Interactions dataset. The study focuses on preparation time, number of ingredients, and nutritional values. After data cleaning and exploratory analysis, hypothesis testing shows that shorter preparation time and fewer ingredients are associated with slightly higher ratings, although the effect sizes are small. Machine learning models are also used to examine whether simple numerical recipe features can predict user ratings. The calorie hypothesis could not be fully tested because a cleaned calorie column was not available in the final dataset. Overall, the results suggest that recipe ratings are influenced by more complex factors beyond basic recipe attributes.

1 Motivation

Online recipe platforms contain large amounts of user-generated data, including ingredients, preparation steps, cooking time, nutrition information, and ratings. These ratings reflect how users respond to different types of recipes. A highly rated recipe may be easier to prepare, simpler, faster, healthier, or more appealing to users.

The motivation of this project is to investigate whether simple measurable recipe features can explain differences in user ratings. Instead of looking at recipes individually, this project uses a large dataset to identify general patterns in how users rate recipes.

The main research question is:

> How do recipe characteristics such as preparation time, number of ingredients, and nutritional values affect recipe ratings?

The main hypotheses are:

- **H1:** Recipes with shorter preparation times receive higher ratings.
- **H2:** Recipes with lower calorie values receive higher ratings.
- **H3:** Recipes with fewer ingredients receive higher ratings.

2 Data Source

The analysis uses the Food.com Recipes and Interactions dataset. The project uses two main files:

- `RAW_recipes.csv`
- `RAW_interactions.csv`

The recipes dataset contains 231,637 recipes, and the interactions dataset contains 1,132,367 user interactions. The recipes file includes recipe-level information such as recipe name, recipe ID, preparation time, tags, nutrition information, number of steps, ingredients, and number of ingredients. The interactions file includes user ratings and reviews.

The two datasets were merged using the recipe identifier. In the recipes file, the identifier is stored as `id`. In the interactions file, it is stored as `recipe_id`. After merging, each user rating could be connected to the corresponding recipe characteristics.

Originally, an external nutrition dataset was planned. However, this file was not available in the final analysis. Although the Food.com recipes file includes a `nutrition` column, the final analysis mainly focuses on preparation time, number of ingredients, and ratings. Although the original proposal planned to enrich the Food.com dataset with an external nutrition dataset, this enrichment could not be completed because the external file was not available in the final project files. Therefore, the final analysis documents this as a limitation and focuses on the reliable variables available in the Food.com dataset.

3 Data Cleaning and Preprocessing

After merging the datasets, missing values and invalid values were handled in the key variables. The main columns used in the analysis were:

- `avg_rating`: average recipe rating calculated from user interactions.
- `minutes`: preparation or cooking time of the recipe.
- `ingredient_count`: number of ingredients in the recipe.
- `ingredient_group`: categorical grouping of recipes into fewer-ingredient and more-ingredient groups.

The following filters were applied:

- `minutes > 0`
- `ingredient_count > 0`

- `minutes <= 300`, to remove unrealistic preparation-time outliers.
- `ingredient_count <= 30`, to remove extreme ingredient-count outliers.
- Rows with missing values in key analysis columns were removed when necessary.

These filters were necessary because some recipes had unrealistic preparation times, including extremely large values. Such outliers could distort summary statistics, visualizations, hypothesis tests, and machine learning results.

An additional `ingredient_count` variable was computed from the ingredients list to represent the number of ingredients in each recipe.

4 Numerical Summary

Table 1 summarizes the cleaned dataset used in the main analysis.

Table 1: Numerical Summary of the Cleaned Dataset

Dataset	n	Mean rating	Median minutes	Median ingredients
Cleaned dataset	231,637	4.3462	40.0	9.0

The cleaned dataset contains 231,637 recipes. The mean rating is 4.3462, which shows that recipe ratings are generally high. The median preparation time is 40 minutes, and the median number of ingredients is 9. These median values were later used as cutoffs in the hypothesis tests.

The high mean rating suggests a positivity bias in the dataset. When most ratings are close to the upper end of the rating scale, it becomes more difficult for simple numerical features to explain rating differences.

5 Exploratory Data Analysis

Exploratory data analysis was used to understand the structure of the dataset and examine possible relationships between recipe features and ratings.

First, the rating distribution was examined. As shown in Figure 1, ratings are concentrated near the upper end of the scale. This indicates that users tend to give positive ratings to recipes.

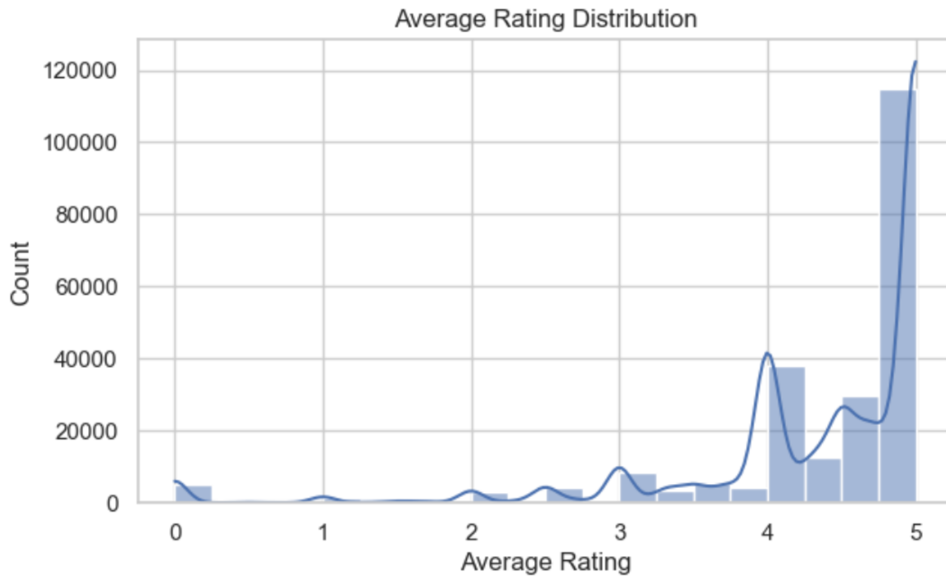

Figure 1: Distribution of average recipe ratings

Second, preparation time was compared with rating using a scatter plot. As shown in Figure 2, the relationship between preparation time and rating appears weak and scattered. This suggests that preparation time alone does not strongly explain recipe ratings.

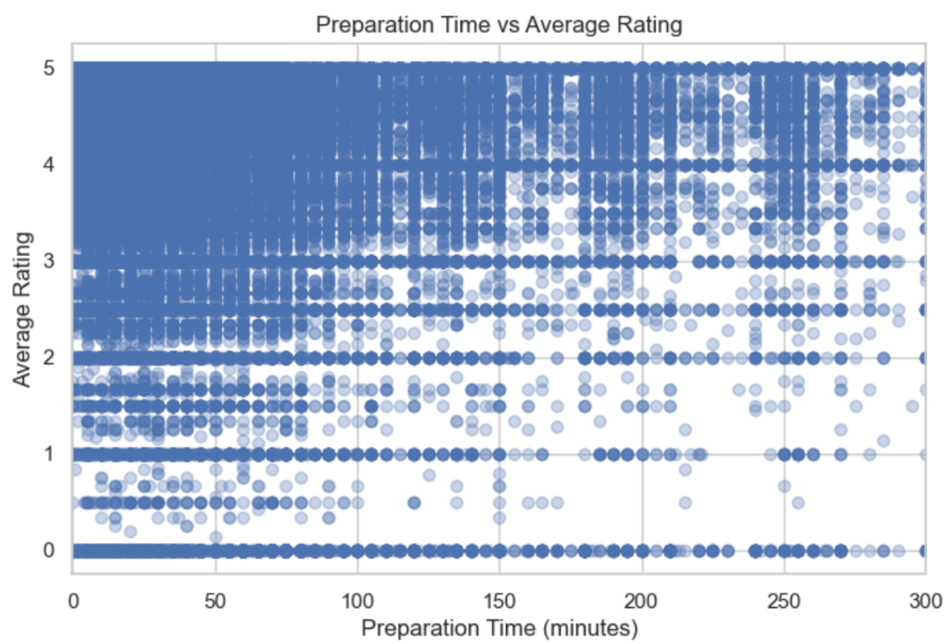

Figure 2: Preparation time versus average recipe rating

Third, ingredient count was compared with average rating. As shown in Figure 3, the relationship between ingredient count and rating is also weak and scattered. This supports the idea that ingredient count alone does not strongly explain rating differences.

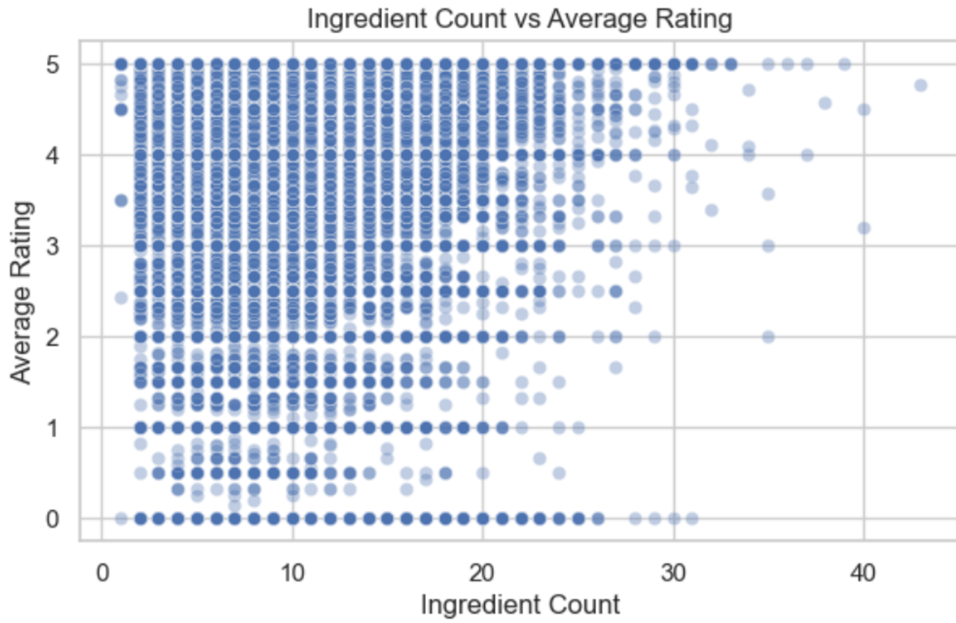

Figure 3: Ingredient count versus average recipe rating

Correlation analysis and grouped averages were also used to examine the relationships between recipe characteristics and ratings. Overall, the visual results suggest that preparation time and ingredient count have weak and scattered relationships with recipe ratings. This supports the later hypothesis testing results: the relationships may be statistically significant, but their practical effect is small.

6 Hypothesis Testing

Independent-samples t-tests were used to test the project hypotheses. Recipes were divided into two groups based on the median value of the tested variable.

For preparation time, the median cutoff was 40 minutes. Recipes with preparation time less than or equal to 40 minutes were compared with recipes above 40 minutes.

For ingredient count, the median cutoff was 9 ingredients. Recipes with ingredient count less than or equal to 9 were compared with recipes above 9 ingredients.

Since the hypotheses predicted a specific direction, one-sided independent-samples t-tests were used.

Table 2: Hypothesis Test Results

Hypothesis	Variable	Median cutoff	Low group mean	High group mean	t-statistic	One-sided p-value
H1	Minutes	40.0	4.3798	4.3047	17.9930	1.2504×10^{-72}
H2	Calories	Not available	–	–	–	Not tested
H3	Ingredients	9.0	4.3509	4.3395	2.7118	0.0033

For H1, the one-sided p-value was 1.2504×10^{-72}, which is below 0.05. Therefore, the null hypothesis was rejected. Recipes with shorter preparation times had a higher mean rating than recipes with longer preparation times.

For H2, the hypothesis could not be evaluated because a cleaned calories column was not available in the final dataset.

For H3, the one-sided p-value was 0.0033, which is also below 0.05. Therefore, the null hypothesis was rejected. Recipes with fewer ingredients had a slightly higher mean rating than recipes with more ingredients.

However, these results should be interpreted carefully. Although H1 and H3 were statistically significant, the actual differences in mean ratings were small. Therefore, preparation time and ingredient count appear to have weak practical effects on recipe rating. These results show statistical association rather than causation, so the analysis does not prove that shorter preparation time or fewer ingredients directly cause higher ratings.

7 Machine Learning

Machine learning was used as an additional analysis step to test whether recipe ratings could be predicted from recipe-level features. The target variable was the recipe rating. The feature columns included preparation time, ingredient count, and other available numerical recipe characteristics.

The dataset was split into training and testing sets using an 80/20 train-test split. Model performance was evaluated using MAE, RMSE, and R^2. MAE and RMSE measure prediction error, while R^2 shows how much of the rating variation is explained by the model.

The models used were:

- Baseline model
- Linear Regression
- Random Forest Regressor

The baseline model predicts the mean rating and provides a simple reference point. This comparison is important because a machine learning model should perform better than a simple baseline to be considered useful.

Table 3: Machine Learning Model Performance

Model	Test MAE	Test RMSE	Test R^2
Baseline	0.5120	0.7192	-0.000004
Linear Regression	0.7938	23.4685	-562.1769
Random Forest	0.6833	0.9878	0.0022

The machine learning results show that the Random Forest model performed better than Linear Regression in terms of R^2. Random Forest achieved a Test MAE of 0.6833, a Test RMSE of 0.9878, and a Test R^2 of 0.0022. However, the baseline model achieved a lower Test MAE of 0.5120 and a lower Test RMSE of 0.7192. This means that the machine learning models did not clearly outperform a simple mean-prediction baseline.

Linear Regression performed poorly, with a very large Test RMSE of 23.4685 and a negative Test R^2 value of -562.1769. This suggests that the linear model was not suitable for this prediction task and produced unstable predictions on the test set. The unusually large RMSE for Linear Regression suggests that the model may have produced extreme predictions for some observations. Therefore, this result should be interpreted as evidence that a simple linear model is unstable for this task.

Overall, the very low R^2 values show that the selected recipe features explain only a very small fraction of the variation in ratings. Therefore, simple recipe characteristics such as preparation time, ingredient count, and other available numerical variables are not sufficient to accurately predict user ratings.

8 Findings

The analysis produced the following key findings:

- The cleaned dataset contains 231,637 recipes, with a mean rating of 4.3462.
- The median preparation time is 40 minutes, and the median ingredient count is 9.
- Shorter preparation time is statistically associated with higher ratings.
- Fewer ingredients are statistically associated with slightly higher ratings.
- The calorie hypothesis could not be fully tested because cleaned calorie data was not available.
- Although H1 and H3 are statistically significant, the actual differences in mean ratings are small.

- Random Forest achieved a slightly positive R^2, but it did not clearly outperform the baseline model in terms of MAE and RMSE.

Overall, the results suggest that shorter and simpler recipes may receive slightly higher ratings, but the practical effect is small. Recipe ratings are likely influenced by more complex factors such as taste, cuisine type, recipe popularity, review sentiment, image quality, and user expectations.

9 Limitations

This project has several limitations.

First, the ratings are user-generated, which means they may contain bias. Users who leave ratings may not represent all users who try a recipe. Also, many ratings are high, which creates positive rating bias and reduces variation in the target variable.

Second, the external nutrition dataset was not available. Because of this, the calorie-based hypothesis could not be fully tested.

Third, the analysis mainly focused on simple numerical features such as preparation time and number of ingredients. These features are easy to measure, but they do not capture the full recipe experience.

Fourth, the machine learning results show that simple numerical variables have very weak predictive power. This suggests that important factors affecting recipe ratings may be missing from the model.

Fifth, the results are correlational. This means that the analysis can identify relationships between variables, but it cannot prove that shorter preparation time or fewer ingredients directly cause higher ratings.

10 Future Work

Future work could improve the project in several ways. First, the `nutrition` column could be parsed into separate variables such as calories, fat, sugar, sodium, and protein. This would allow a more complete test of the nutrition-related hypothesis.

Second, text analysis could be added using recipe tags, descriptions, preparation steps, and user reviews. These text-based features may capture information that simple numerical variables miss.

Third, richer machine learning models could be tested using categorical variables such as cuisine type, recipe tags, and recipe category. The analysis could also be repeated separately for different recipe categories, such as desserts, soups, main dishes, and healthy recipes.

11 Web Application

As an additional part of the project, a minimal web application was developed to make the recipe dataset easier to explore. The web app allows users to search recipes by keyword and filter them by minimum rating, preparation time, and number of ingredients. For example, a user can search for highly rated recipes that take less than 30 minutes and use fewer than 10 ingredients.

This web app makes the analysis more practical because users do not need to inspect the full dataset or notebook. Instead, they can interact with the results through simple filters, recipe cards, and detail pages showing ingredients and preparation steps. The app mainly serves as an interactive extension of the analysis rather than a full recipe recommendation system.

12 Takeaways

- Shorter preparation time and fewer ingredients are statistically significant, but the actual effect size is small.
- Simple numerical features such as preparation time and ingredient count are not enough to accurately predict recipe ratings; the machine learning models did not clearly outperform the baseline.
- The implemented web app makes the project more useful by allowing users to search and filter recipes based on rating, preparation time, and ingredient count.

13 Conclusion

This project analyzed the relationship between recipe characteristics and user ratings using the Food.com Recipes and Interactions dataset. The main focus was on preparation time, number of ingredients, and nutritional values.

The analysis showed that shorter preparation times and fewer ingredients are statistically associated with slightly higher ratings. However, the differences are small. This means that the results are statistically significant but weak in practical terms.

The machine learning results support the same conclusion. Random Forest performed better than Linear Regression in terms of R^2, but it still achieved a very low Test R^2 value of 0.0022 and did not clearly outperform the baseline in terms of prediction error.

The implemented web application also makes the project more accessible by allowing users to explore recipes through simple search and filtering tools.

The main conclusion is that recipe ratings are influenced by more complex factors than preparation time and ingredient count alone. While simple recipe characteristics provide some insight into user preferences, future work should include richer nutritional, textual, and categorical features to better understand what makes a recipe highly rated.

14 AI Assistance Disclosure

AI assistance was used as a writing and editing support tool during the preparation of this final report. It helped with organizing the report structure, improving wording, increasing clarity, and revising LaTeX formatting. The data analysis, code execution, notebook outputs, hypothesis test results, machine learning results, web application, and final conclusions were based on my project files and outputs. I reviewed and verified the final report before submission.

15 References

- Food.com Recipes and Interactions Dataset. Kaggle dataset used for recipe metadata, ingredients, preparation time, user ratings, and user interactions.